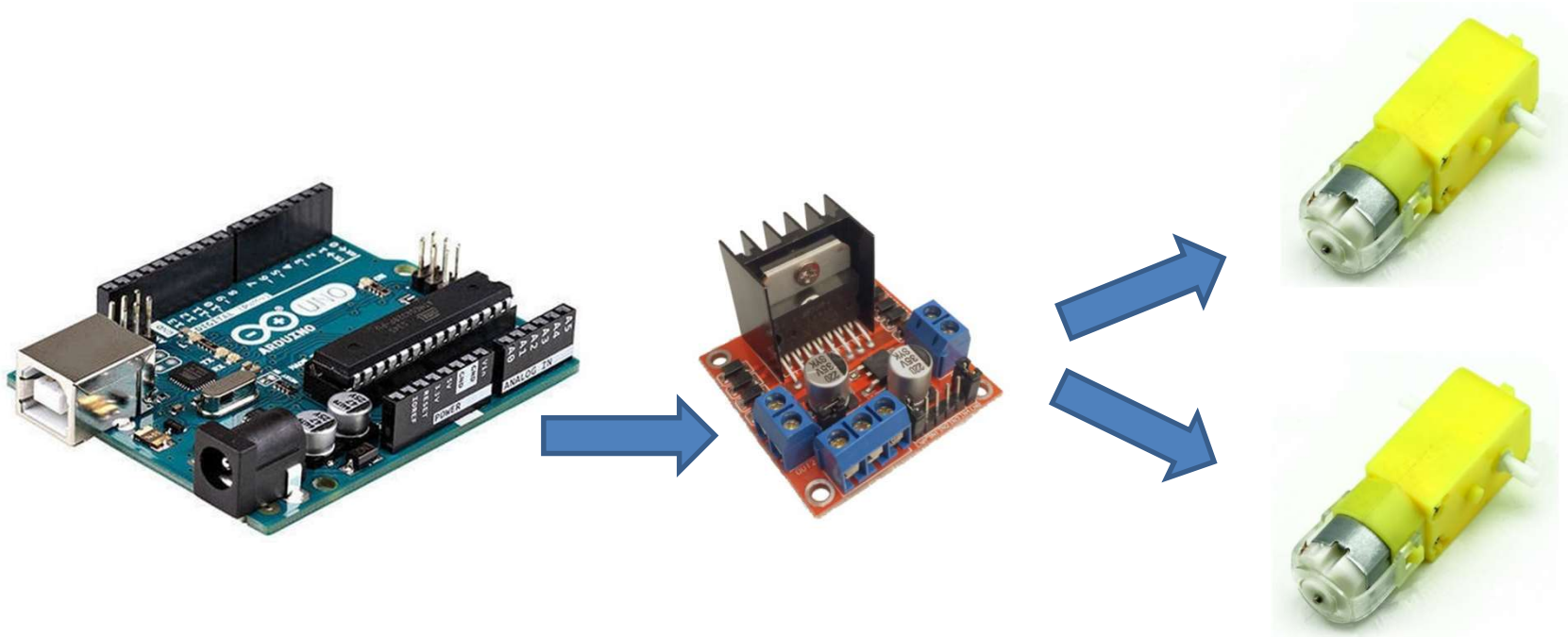
Driver L298: Acionando motores DC com Arduino



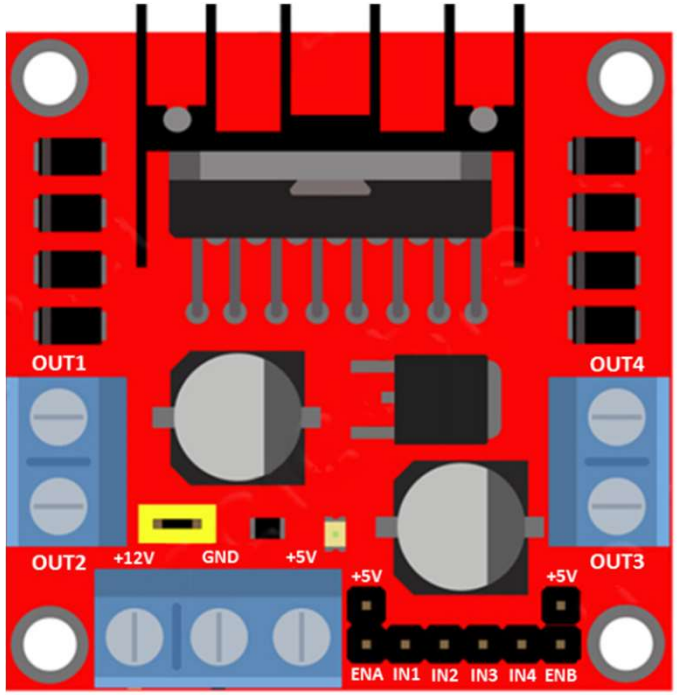
Driver L298: Acionando motores DC com Arduino

Objetivos:

- Implementar o firmware de controle
- Montar o circuito
- Efetuar testes
- Implementar função ou classe para controle de direção do robô
- Iniciar a implementação do robô físico

Driver L298: Acionando motores DC com Arduino

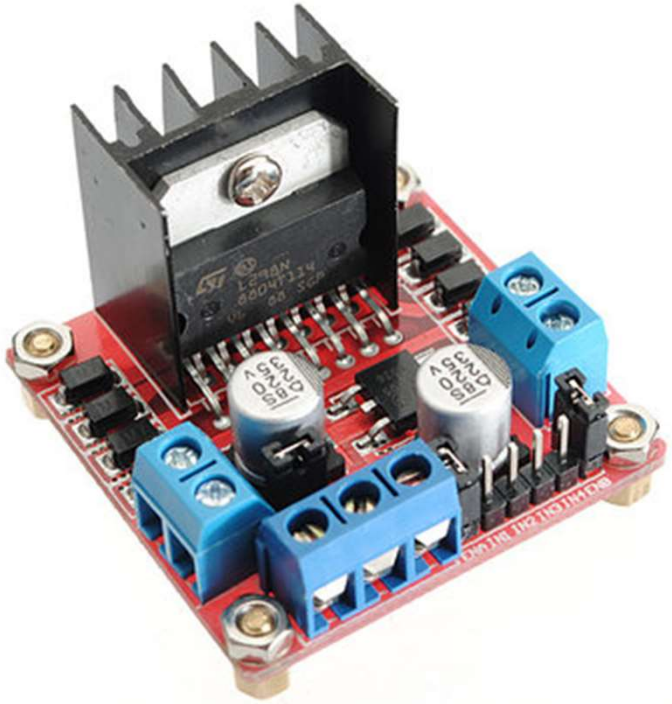
Pinagem: L298



Entradas de Controle

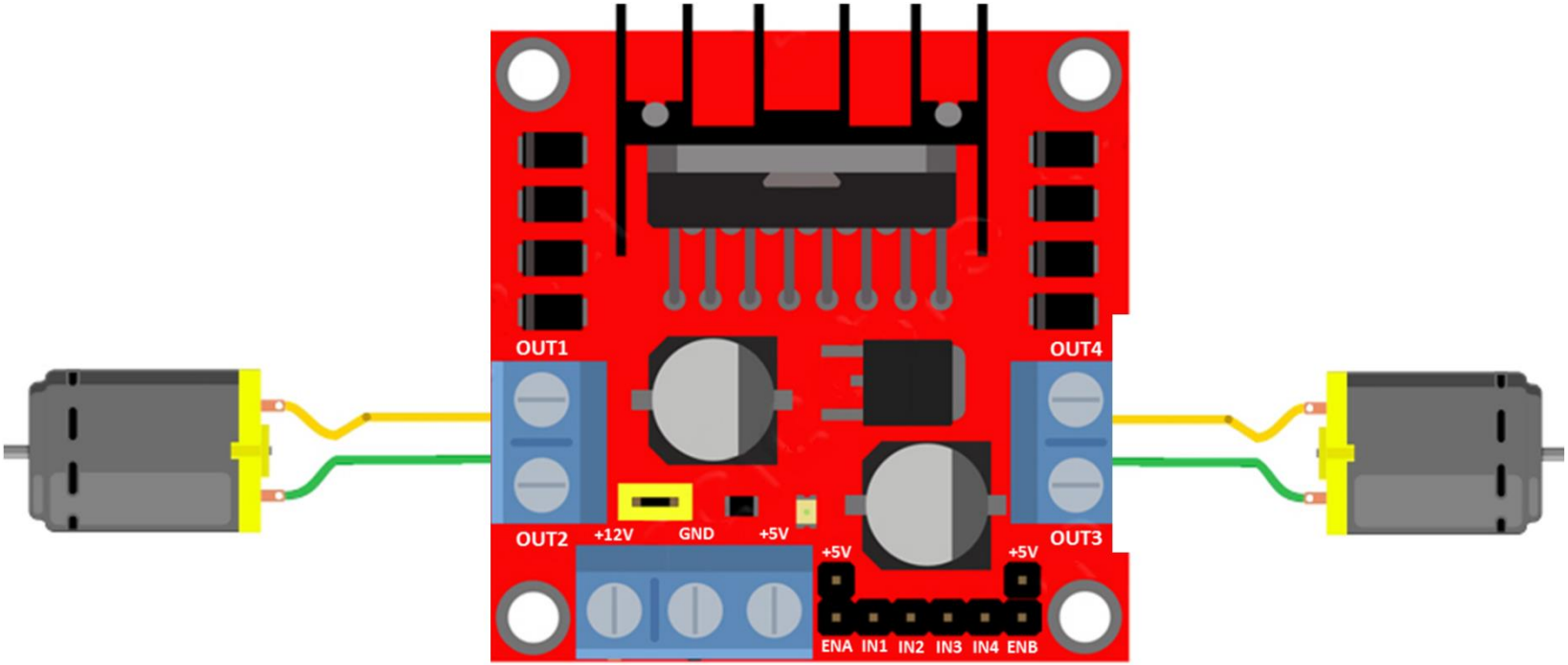
Driver L298: Acionando motores DC com Arduino

Pinagem: L298



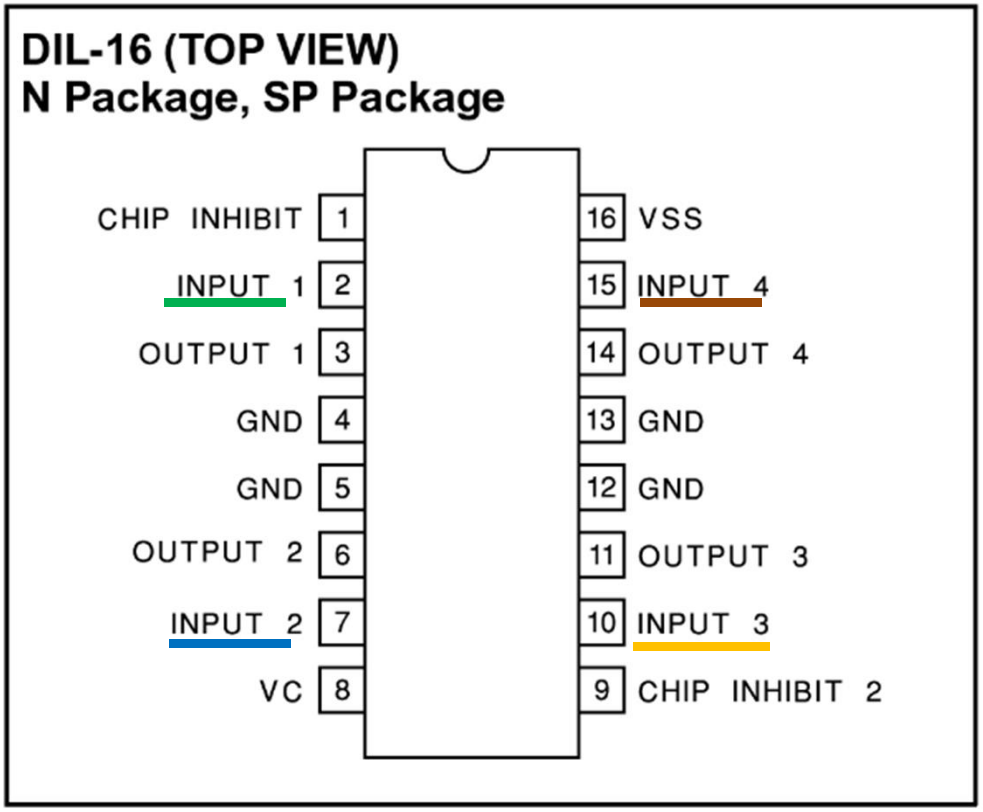
Driver L298: Acionando motores DC com Arduino

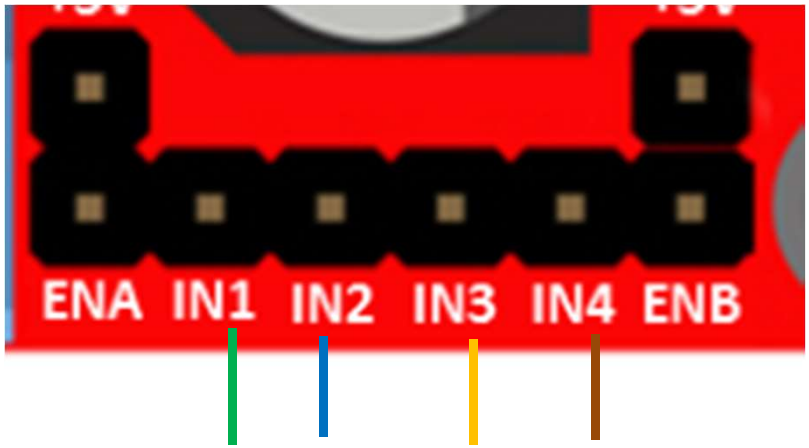
Pinagem: L298



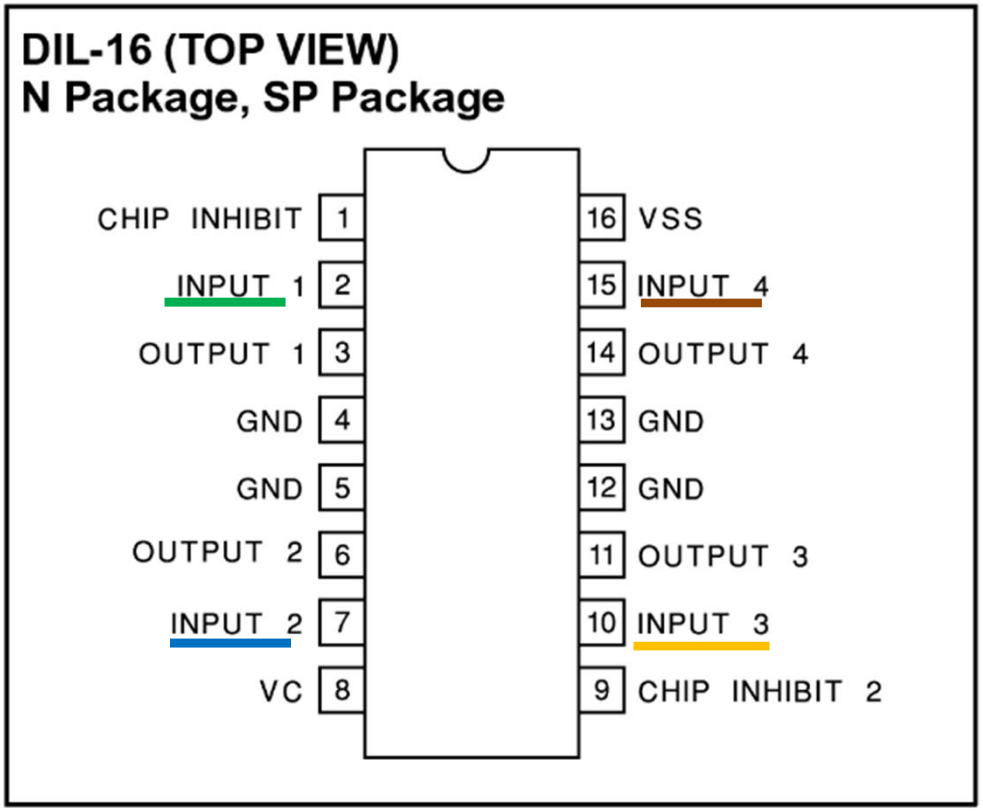
Driver L298: Acionando motores DC com Arduino

Pinagem: L293 relacionamento com L298





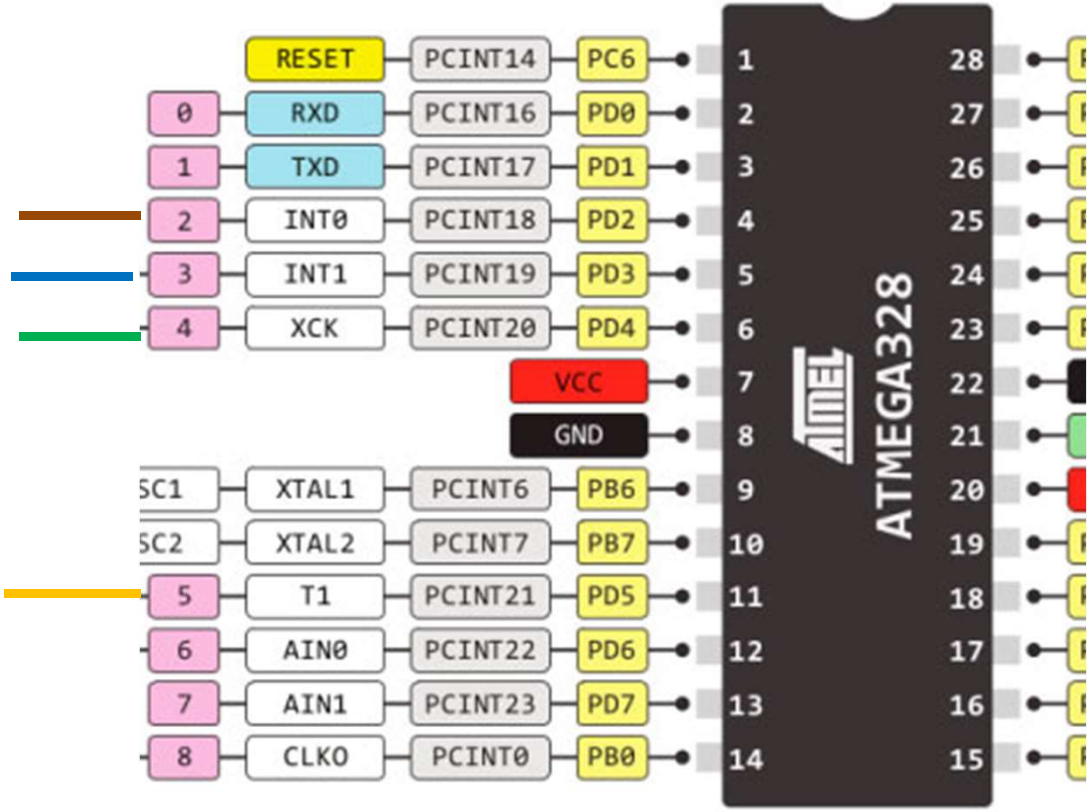
Driver L298: Acionando motores DC com Arduino
Simulação com L293 no Tinkercad

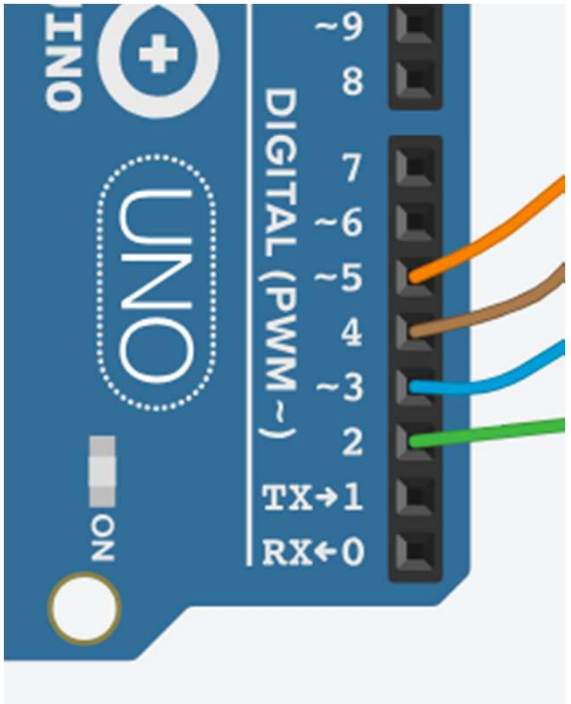


Pin 2/Input 1	Pin 7/Input 2	Motor Connected between Pins 3 and 6)Function
N/A	N/A	Motor Stops
HIGH	HIGH	Motor Stops
LOW	LOW	Motor Stops
HIGH	LOW	Motor Turns Anti-Clockwise
LOW	HIGH	Motor Turns Clockwise

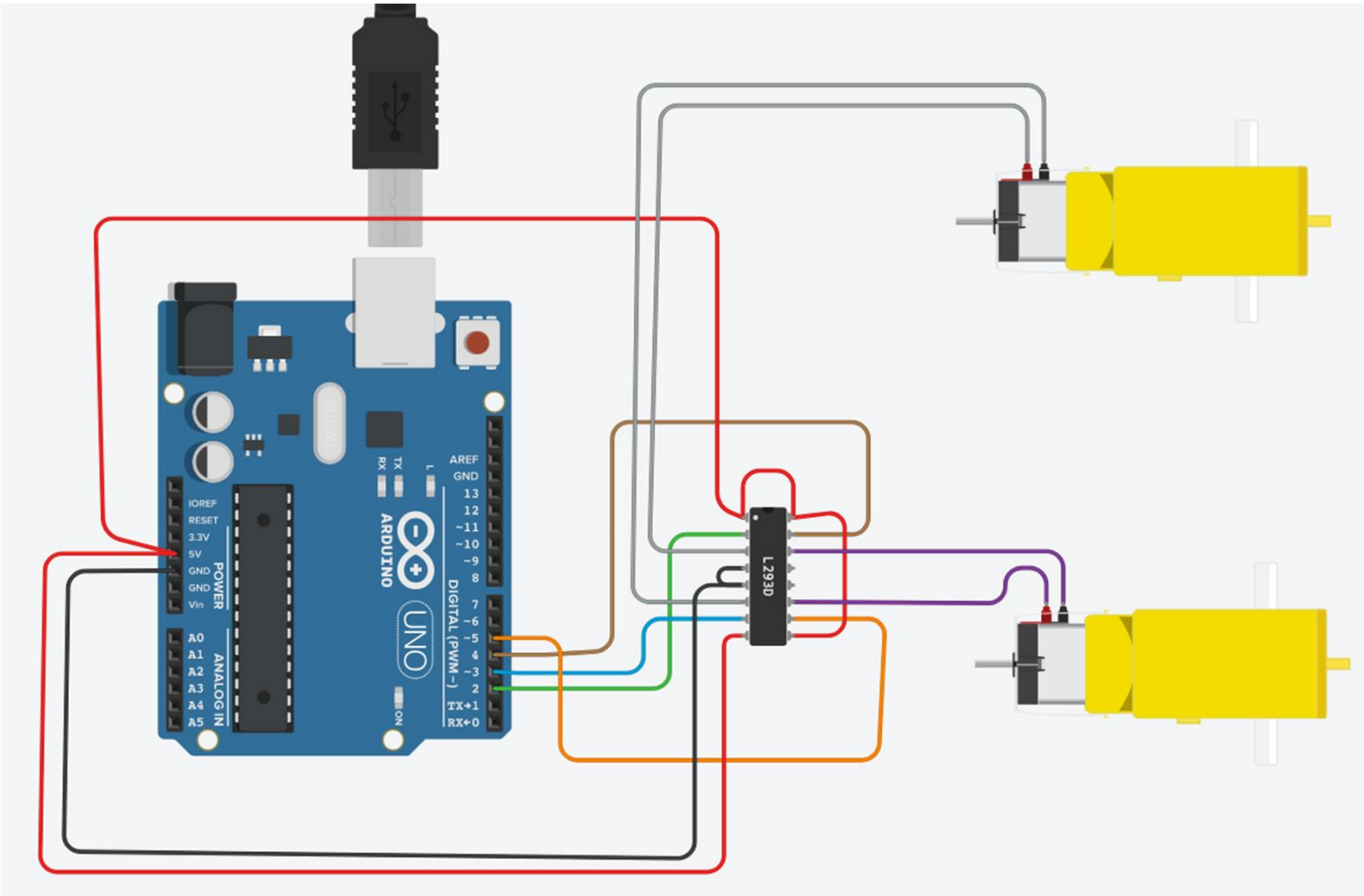
Pin 10/Input 3	Pin 15/Input 4	Motor Connected between Pins 11 and 14) Function
N/A	N/A	Motor Stops
HIGH	HIGH	Motor Stops
LOW	LOW	Motor Stops
HIGH	LOW	Motor Turns Anti-Clockwise
LOW	HIGH	Motor Turns Clockwise

Driver L298: Acionando motores DC com Arduino

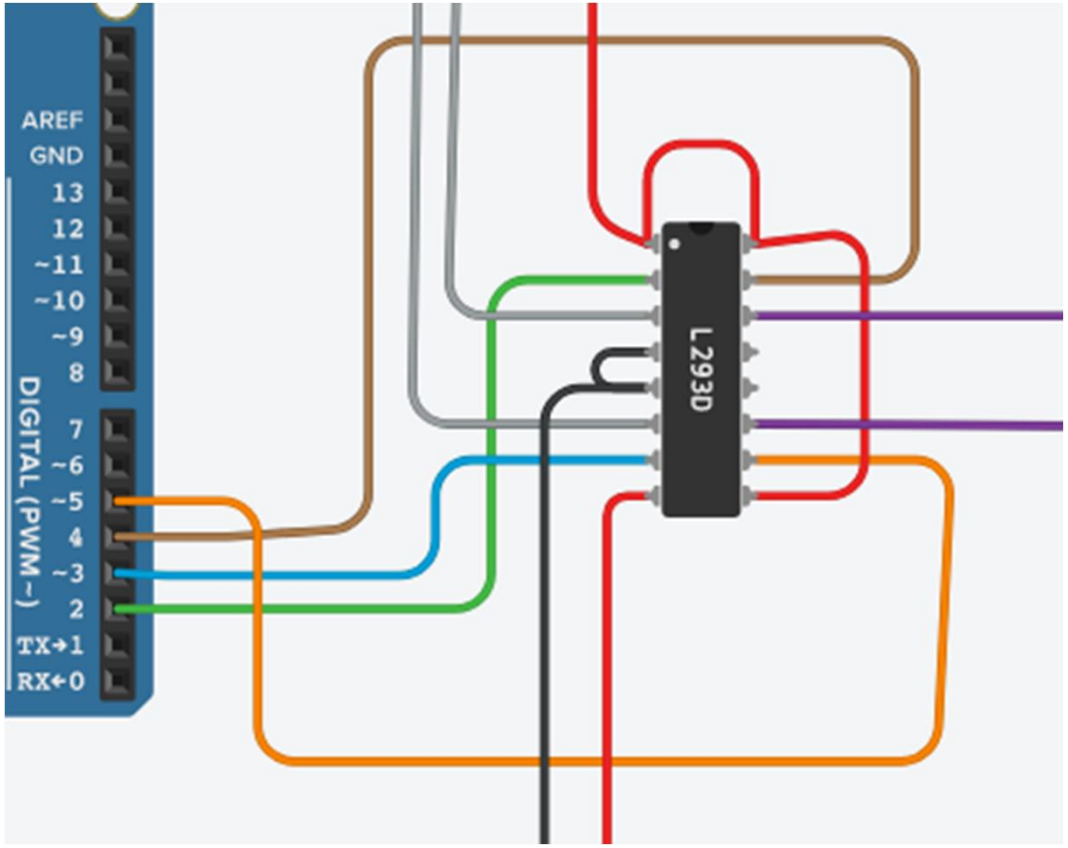




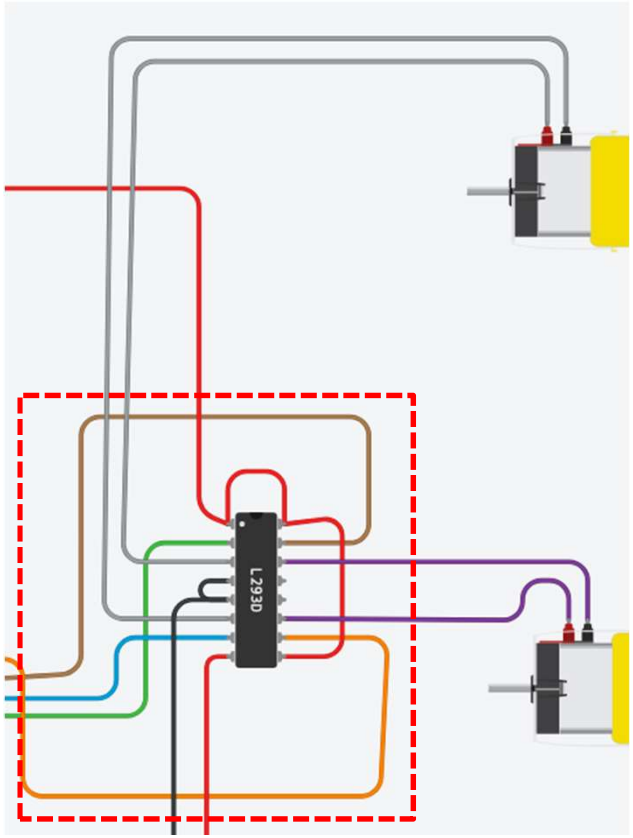
Driver L298: Acionando motores DC com Arduino



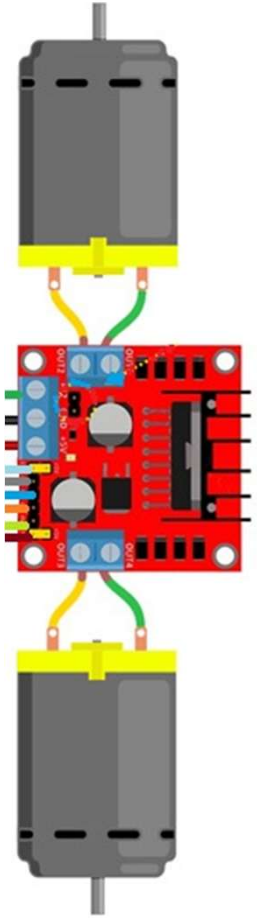
Driver L298: Acionando motores DC com Arduino



Driver L298: Acionando motores DC com Arduino



equivalente



Driver L298: Acionando motores DC com Arduino

13.4.9 DDRD – The Port D Data Direction Register

Bit	7	6	5	4	3	2	1	0	
0x0A (0x2A)	DDD7	DDD6	DDD5	DDD4	DDD3	DDD2	DDD1	DDD0	DDRD
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

```
int main (void)
{
 Serial.begin(9600);
 DDRD = B00111100;
```

Driver L298: Acionando motores DC com Arduino

13.4.8 PORTD – The Port D Data Register

Bit	7	6	5	4	3	2	1	0	
0x0B (0x2B)	PORTD7	PORTD6	PORTD5	PORTD4	PORTD3	PORTD2	PORTD1	PORTD0	PORTD
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

```
while(1) {
PORTD = B00101000;
_delay_ms(3000);
}
```

Driver L298: Acionando motores DC com Arduino

Pin 2/Input 1	Pin 7/Input 2	Motor Connected between Pins 3 and 6)Function
N/A	N/A	Motor Stops
HIGH	HIGH	Motor Stops
LOW	LOW	Motor Stops
HIGH	LOW	Motor Turns Anti-Clockwise
LOW	HIGH	Motor Turns Clockwise

Pin 10/Input 3	Pin 15/Input 4	Motor Connected between Pins 11 and 14) Function
N/A	N/A	Motor Stops
HIGH	HIGH	Motor Stops
LOW	LOW	Motor Stops
HIGH	LOW	Motor Turns Anti-Clockwise
LOW	HIGH	Motor Turns Clockwise

```
while(1) {
PORTD = B00101000;
_delay_ms(3000);
}
```

Driver L298: Acionando motores DC com Arduino

Próximos Passos:

- Acrescentar comandos de controle para acionar e alterar, direção
- Implementar função ou classe para controle de direção do robô
- Iniciar a implementação do robô físico